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SCIENCE ET JEUNESSE
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Physique/Technique

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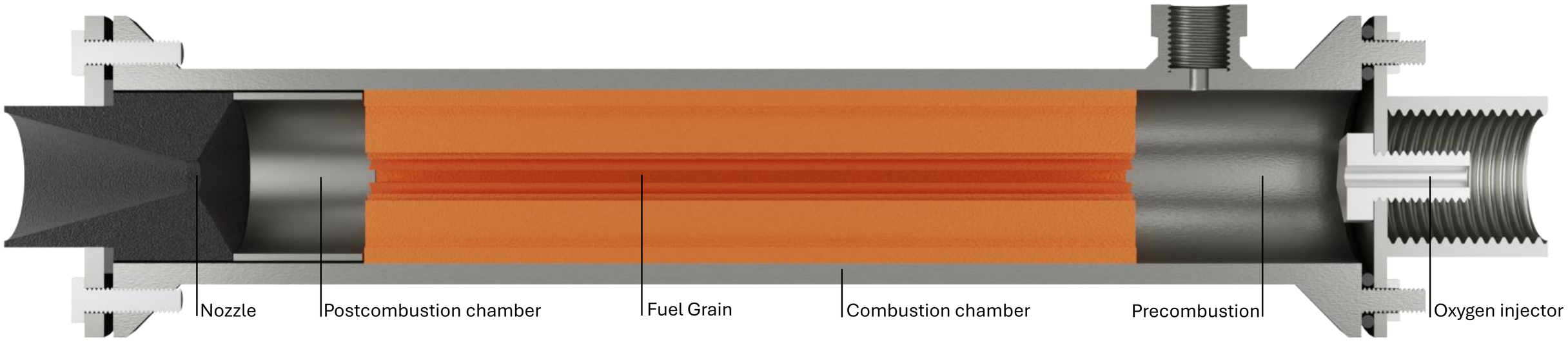
Sacha Bumann, 02.01.2008
Samuel Gebhard, 08.02.2007

GYMNASE DE MORGES

Design and construction of a

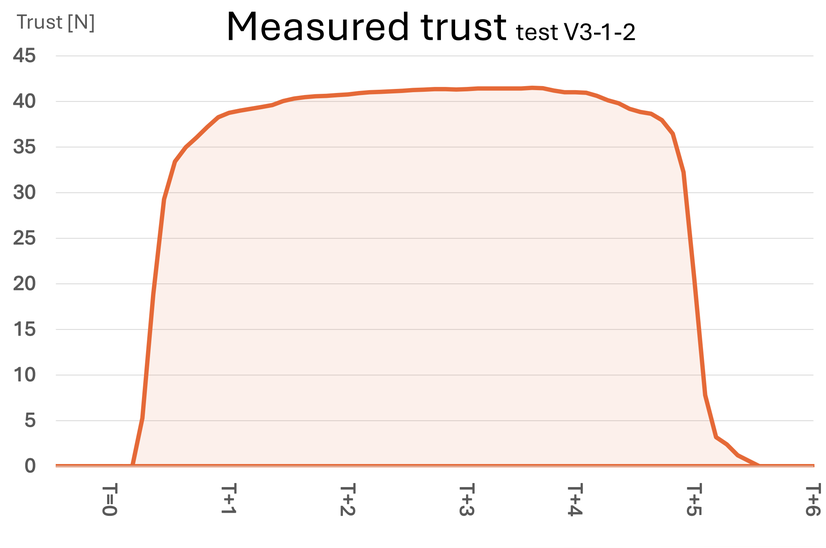
HYBRID ROCKET ENGINE

Demonstrator



INTRODUCTION

Hybrid propulsion combines a liquid oxidizer with a solid fuel, offering a smart balance between efficiency, control, and simplicity. Unlike complex bi-liquid engines, hybrids are more accessible while outperforming solid motors. We designed and tested an experimental engine to evaluate its viability in model rocketry. The goal was to achieve a reliable, high-performance engine with limited resources. Results confirm its potential as a reliable and effective solution for small-scale launches.



43N

The reliable and stable max thrust we achieved during fixed duration tests of 5s.

DEVELOPMENT

- ❑ Design : engine conception using 3D modeling software and combustion simulations.
- ❑ Construction : combustion chamber out of machined steel and nozzle out of graphite.
- ❑ Optimization : collected and analyzed sensors data from multiple successful test firings, applied an iterative design approach to it, refining the system after each test to boost performance and reliability.
- ❑ Testing stand : design and build a custom stand to maintain the engine during tests and collect sensor data as precisely as we could.

RESULTS

- ❑ 40 newtons stable average thrust during 5-second burns with peak at 43 newtons.
- ❑ Constant and stable pressure inside the combustion chamber during the tests.
- ❑ 16 out of 18 tests successful, proving system reliability.
- ❑ Temperature remained controlled below 150°C showing no risk to the engine.
- ❑ Results show that hybrid propulsion is a viable option for model rocketry and that it is possible to build an efficient rocket engine with relatively low resources and within 9 months

This work has given us the opportunity to explore and better understand rocket propulsion.

In total we achieved **18** concluded tests

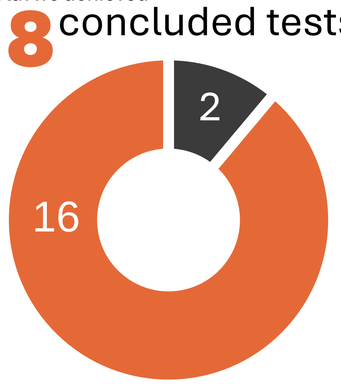




Image taken during the test V3-1-2 by using a soldering mask to make appear the “shock diamonds”, proving that the gas exit speed is greater than the speed of sound.